Industrial Structure, Technological Change, and U.S. Wage Inequality

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February 2026

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Background

- Wage inequality has been on the rise in the United States over the last decades
  - what are the key drivers behind rising inequality?

① Employment polarization and Skill-Biased Technological Change {*e.g.*, Autor *et al.* (2006)}

- *within-firm* dimension $\rightarrow$ *skill*-premium {*e.g.*, Katz and Autor (1999), Lemieux (2010)}
- huge relative supply of skilled workers $\rightarrow$ SBTC {*e.g.*, Tinbergen (1974), Card and DiNardo (2002)}

② Room for supply-side {*e.g.*, Krueger and Summers (1987), Leonardi (2007), Card *et al.* (2013), Song *et al.* (2019)}

- *between-industry* dispersion drives the overall change in U.S. wage inequality {*e.g.*, Haltiwanger *et al.* (2023, 2024); Briskar *et al.* (2022) for Italy}
  ▸ Graph
- types of industries shape the downward polarization of employment {*e.g.*, Cerina *et al.* (2021)}

This paper: central intuition

- Thorough understanding of wage inequality by re-framing the SBTC view
  - labour market features $\rightarrow$ task-based employment {*e.g.*, Autor and Dorn (2013), Cossu *et al.* (2024)}
  - sectoral organization of production $\rightarrow$ evolving structural complementarities

- Inequality in wages can be influenced along two dimensions
  - if elasticities of substitution among factors of production are uniform across industries, mainly differences in capital and labour accumulation (SBTC) $\leftrightarrow$ ***quantity effect***
  - otherwise, similar inputs growth lead to diverging wage outcomes $\leftrightarrow$ ***structural effect***

☞ Technological changes on substitution elasticities is what matters for inequality
  - industry-level technological heterogeneity rather than factor accumulation alone

What's to come

Challenge: to address the determinants of real wage variance

Capture: dominant role of heterogeneity in the U.S. industry dimension

$\rightarrow$ disentanglement between a *quantity* and a *structural* effects

How:
① empirical motivation at 3-digit U.S. 2017 NAICS level
② GE model: endogenous sorting and technological differences
③ structural estimation and counterfactual analysis

Question

What drives the industry-heterogeneous trend in wages?

The paper in one slide

① Empirics: industries contributing more to wage inequality are those ▸ Stylized facts

- significantly adopting ICT in physical capital, and where ...

  ... the substitution of routine with non-routine workers is more pronounced.

② Model ▸ Model overview

- workers' sorting and segregation effects + wage premium from SBTC;
- it is able to address the observed between-industry wage inequality.

③ Results: importance of *structural sectoral heterogeneity* on U.S. wage inequality ▸ Take-aways

- structural effects dominate quantity effects $\rightarrow$ 94% vs. 6-15%, respectively;
- substitution across job tasks is the dominant channel;
- labour market concentration magnifies structural effects, up to 98%;
- not primarily driven by aggregate monopsony power.

Related literature

- Wage inequality in U.S. due to industrial composition {*e.g.*, Slitcher (1950), Cullen (1956), Krueger and Summers (1988), Caselli (1999), Allen (2001), Haltiwanger *et al.* (2023, 2024), Card *et al.* (2024)}
  - → ***Contribution***: incorporate *structural* differences among U.S. industries on inequality
- Task-based literature {*e.g.*, Autor *et al.* (2006), Goos *et al.* (2009), Acemoglou and Autor (2011), Cortes *et al.* (2017), Cerina *et al.* (2021), Siena and Zago (2024), Cossu *et al.* (2024)}
  - → ***Contribution***: study how the substitutability across tasks affects inequality
- Elasticity of substitution between capital and labour. In particular:
  - (*i*) secular decline in the labour share and role of industries {*e.g.*, Glover and Short (2023)}
  - (*ii*) structural transformation {*e.g.*, Buera and Kaboski (2012), Herrendorf *et al.* (2015)}
  - → ***Contribution***: estimated industry-level capital-labour substitution elasticities < 1

Motivating Evidence

U.S. industries digitalization and labour force composition

- The `BEA Detailed Data for Fixed Assets` provides data on:
  - the stock of 96 different types of capital, classified in *intangible* capital, *digital equipment* and structures, and *physical* capital

▸ BEA construction

- The `BLS Occupational Employment and Wage Statistics` has data on:
  - more than 100 occupations according to the Standard Occupational Classification (SOC), classified in *routine* and *non-routine* tasks

▸ BLS construction

- *Unit*: 62 private 3-digits U.S. 2017 NAICS industries
- *Period*: 2003-2022, annual

Micro-econometrics methodology: panel data regressions, variance decompositions, shift-share

Labour force composition: substitution or size?

How much of the changing share of task-a is due to changing sizes of industries and how much is due to changes in workforce composition within those industries?

$$\Delta\left(\frac{\ell(a)}{\ell(a')}\right) = \underbrace{\sum_s \overline{\left(\frac{\ell(a,s)}{\ell(s)}\right)}\Delta\left(\frac{\ell(a,s)}{\ell(a',s)}\right)}_{\textit{within-industry: substitution effect}} + \underbrace{\sum_s \overline{\left(\frac{\ell(a,s)}{\ell(a',s)}\right)}\Delta\left(\frac{\ell(a,s)}{\ell(s)}\right)}_{\textit{between-industry: size effect}}$$

Table 1: Labour force changes

	routine		***non-routine***	
Interval	*within*	*between*	*within*	*between*
2003-2008	83%	17%	38%	62%
2009-2015	81%	19%	58%	42%
2016-2022	78%	22%	69%	31%

▸ Figure (2-digits) ▸ Shares (3-digits)

Table 2: Combined regressions by groups, percentage changes

	$\Delta \log w(s)$			
	bottom	*middle-bottom*	*middle-top*	*top*
$\beta_{\Delta k}$	-.693	.302***	.078	-.082*
	(.618)	(.072)	(.179)	(.042)
$\beta_{\Delta \ell}$	.041***	-.297	.015***	.052*
	(.004)	(.247)	(.003)	(.025)
$\beta_{\Delta k \times \Delta \ell}$	.876***	-1.304	-.643	1.560*
	(.100)	(2.945)	(.470)	(.789)

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022. Each Fixed Effects (FE) regression – industries by their overall growth in real wage –, is of the form* $\Delta \log w_{\omega t}(s) = \beta_{c,\omega} + \beta_{i,\omega} \mathcal{X}_{i,\omega t} + \delta_{j,\omega} \mathcal{Z}_{j,\omega t} + u_{\omega t}$*, with* $\mathcal{X}_i$ *representing the percentage change in both ICT-to-physical capital and non-routine-over-routine workers ratios, and* $\mathcal{Z}_j$ *being a set of time-varying controls. Variables are all in log format. Constant not reported to save space. Source: BEA, BLS and own calculations.*

▸ Correlations ▸ Quantiles, trend ▸ Regressions, changes

Possible interpretations

- The facts so far are proceeded by two more:
  - the tails of the wage growth distribution mainly contributes to inequality; ▸ Fact 1: Contribution
  - technological differences are related to ICT and not to physical capital. ▸ Fact 2: Capital gaps

- Not only a matter of isolated changes, but rather pivotal appears the interaction
  → in industries contributing more (Fact 1), substitutability across worker-types (Fact 3), . . .
  . . . interacting with technological gaps (Fact 2), drives real wage dispersion (Fact 4)

- Wage inequality across industries could be the result of
  - a ***quantity*** effect, under different paths in factor of productions;
  - a ***structural*** effect, due to changes in technology parameters governing the relationship between capital and labour types;

  ☞ these general equilibrium issues will be addressed through the estimated model.

Model

Households

- Unit mass of households, each labelled as i, split in types $a \in \{rt, nrt\}$ ‣ Intertemporal utility

$$\mathcal{U}^{i}_{h}(a,s) = log\, \mathcal{C}^{i} + log\, \mathcal{B}_{h}(a,s) + \wp^{i}_{h}(a,s)$$

  - each chooses consumption, invests in both capital types, and receives dividends

- Idiosyncratic productivity when working as type-a for firm-h in industry-s ‣ Indirect utility
  - drawn once from a *bivariate Frechét-type distribution* ... ‣ Frechét variability

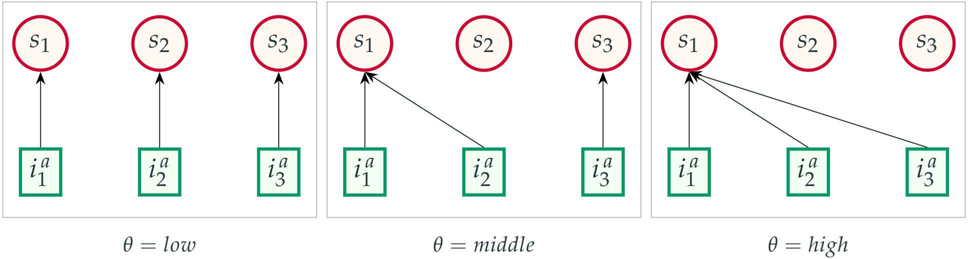
    ... whose shape parameter identifies common productivity dispersion $\longleftrightarrow$ θ

- Analytical form for the measure of each worker-type in firm h, s ‣ Validation: employment

$$\ell_{h}(a,s) = \left(\frac{w_{h}(a,s)}{W(a)} \frac{\mathcal{B}_{h}(a,s)}{\mathcal{B}(a)} \right)^{\theta}$$

→ **Remark:** *sorting* and *segregation* effects in interaction with the *wage premium*

Productivity dispersion as a proxy of sorting and segregation



- ✫ workers in the same task are "highly rival factors" (from Hicks 1932)

Market structure

- Three layers: (i) *final aggregator*, (ii) *sectoral bundler*, and (iii) *firms* ▸ Output aggregators

- *Technology* $\longrightarrow$ Cobb Douglas-Nested CES ▸ Production function

CES-1:	non-routine labour is complementary to ICT capital	$\longleftrightarrow$	ρ
CES-2:	routine labour is substitutable with *CES-1*	$\longleftrightarrow$	σ
C-D:	non-ICT capital together with *CES-2* part	$\longleftrightarrow$	α

- Differentiated wages for *routine* and *non-routine* workers but at industry level, due to
  - flexible movement of workers across firms; ▸ Proposition 1
  - *max-stability property* of the *Frechét* distribution $\leftrightarrow$ once choosing a workplace, a worker will not move across industries ▸ Remark 1

▸ Firm problem ▸ Wages ▸ Aggregation ▸ Equilibrium ▸ Validation: wage

Discussing the model

- Industry-level wage premium is determined by its specific characteristics:
  - endowments and accumulation of factors of production $\leftrightarrow$ ***quantity*** effect;
  - two-sided technological combination across capital and labour types $\leftrightarrow$ ***structural*** effect;
  - $\sigma > \rho \rightarrow$ capital-task complementarity $\rightarrow$ strongest role of SBTC. {*e.g.*, Krusell *et al.* (2000)}

- Endogenous Roy (1951)-style sorting of workers into workplaces
  - workers are themselves of varying capacities depending on where they are employed in;
  - structural distortion, as ***labour market concentration*** is endogenously determined.

- Wage inequality is the outcome of industry-level and labour market characteristics
  - ☞ it is then possible to design counterfactuals to disentangle its key drivers

from Theory to Data

Calibration strategy

- Vector of parameters to calibrate: $\Theta = \left(\alpha_s,\, \epsilon,\, \lambda_s,\, \mu_s,\, \theta,\, \rho_s,\, \sigma_s\right), \forall s \in \{bot, mid, top\}$

  ① *data and external calibration*. In particular:

  ⋆ industries' shares of non-ICT capital, α_s, directly from BEA tables, and offline calibration for ϵ;

  ② *estimating equations* for elasticity of substitution parameters, ρ_s, σ_s;

  ③ *Method of Simulated Moments* (MSM) for distributive weights parameters, λ_s, μ_s, and degree of labour market concentration, θ. In particular:

  ▸ MSM targets ▸ MSM estimates

  ⋆ $\lambda_s \longleftrightarrow$ industry-specific ICT capital in the aggregate stock;

  ⋆ $\mu_s \longleftrightarrow$ routine workers in the data with that predicted by the model in eq. (10);

  ⋆ $\theta \longleftrightarrow$ routine real *log*-wage premium of top relative to bottom groups.

→ **Model fit**: perfect targeting of observed wage inequality values

▸ Variances

Elasticities of substitution

- A shadow implication of the *data* section is the divergence in elasticities of substitution
  - ☞ identification: ***negative relationship between labour share and relative ICT stock*** ▸ Figure
- Estimating equations from the inter-temporal GE properties of the model: ▸ Derivation

$$\frac{s_\ell(nrt,s)}{1-s_\ell(nrt,s)}\widehat{s}_\ell(nrt,s) = \beta_c + (\rho-1)\,\widehat{\zeta}\,(s) + u \tag{8.1}$$

$$\frac{s_\ell(s)}{1-s_\ell(s)}\widehat{s}_\ell(s) = \beta_c + (\sigma-1)\,\widehat{\zeta}\,(s) + \beta_k\left(\widehat{\frac{\ell(nrt,s)}{k(ict,s)}}\right) + u \tag{8.2}$$

  - relationship between trends in labour shares and trends in relative capital quantities
  - negative when estimated elasticities imply sufficient complementarity, $\rho, \sigma < 1$ ▸ Estimates
  - estimation through panel robust regression techniques (*i.e.*, down-weighting outliers)

Calibration

Table 3: Summary of calibration

	parameter	value				source
		bottom	*middle*	*top*	*global*	
α	*physical capital, share of $y(s)$*	0.263	0.195	0.514		*data*
ϵ	*demand elasticity across firms*				6	*external*
μ	*weight of routine workers in $y(s)$*	0.676	0.495	0.337		*MSM*
λ	*ICT capital share in $Q(s)$*	0.457	0.465	0.451		*MSM*
θ	*households' productivities dispersion*				11.3	*MSM*
ρ	*EoS, ICT capital and non-routine*	0.329	0.420	0.249		*estimation*
σ	*EoS, routine and ICT composite*	0.634	0.400	0.766		*estimation*

Parameters by industry groups. Under "data" the values are directly computed from the data sources, while "external" sets standard values from the literature. "MSM" refers to the Methods of Simulated Moments, and "estimation" reports the previously estimated values from Table C.3.

▸ Model fit ▸ Untargeted ▸ Quantiles

Steady-state discussion: real log-wage variances

Table 4: Implied fit of wage variances

moment	*data*	*model*
routine wage variance, across industries	2.285	2.289
non-routine wage variance, across industries	2.314	2.311
between-industry wage variance	2.299	2.300

Variances computed according to eq. (A.1) over the period 2003-2022. The first two rows compute the measure for routine and non-routine tasks separately, while the last row directly reports the wage dispersion across industries.

☞ the model built target all the relevant measures of wage inequality

Counterfactual analysis

→ What's next? *Counterfactual analysis*

- contribution of variations in parameters and/or series on *between-industry* wage inequality

- Either for total industry employment, and for routines and non-routines separately:
  1. changing one or more key parameters $(\rho_s, \sigma_s, \theta)$ at a time, fixing the others $(\alpha_s, \mu_s, \lambda_s)$;
     - ⋆ Tables C.8, C.10, C.12
  2. keeping fixed the key parameters, while changing the others;
     - ⋆ Tables C.9, C.11, C.13
  3. remove heterogeneous parameters, and vary only capital, labour, and productivity series;
     - ⋆ Tables C.17, C.18

▸ Productivity, mean ▸ Productivity, new

  4. no changes in parameters, and let to vary only capital and labour series.
     - ⋆ Tables C.14, C.15, C.16

▸ MSM re-estimates ▸ Models for counterfactuals

Which parameters to pick?

- Selected parameters:

$$\begin{array}{rlcl} (i) & \text{trend-differences in structural transformations} & \longleftrightarrow & \rho \text{ and } \sigma; \\ (ii) & \text{shifts in labour market concentration} & \longleftrightarrow & \theta. \end{array}$$

- Relevance of the changes occurred in the two elasticities of substitution . . .
  - bottom $\longrightarrow$ $\rho \uparrow\uparrow$ and $\sigma \downarrow$;
  - middle $\longrightarrow$ $\rho \downarrow$ and $\sigma \uparrow$;
  - top $\longrightarrow$ $\rho \uparrow$ and $\sigma \downarrow$.

. . . considering two separate time windows, 2003-2012 and 2013-2022.

▸ Time-varying estimates

Increasing sorting and segregation effects, captured by θ

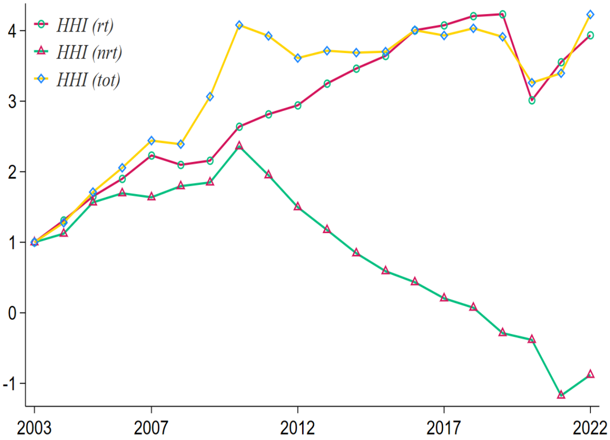
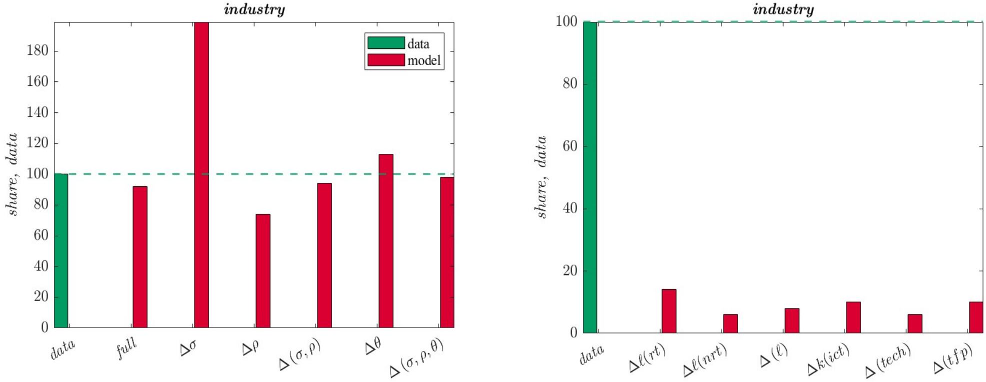


Figure 2: Labour market concentration

HHI
HHI industries

Model re-calibration counterfactuals



(a) Changing parameters

(b) Changing series, unique calibration

Figure 3: Impact on between-industry wage inequality

Conclusions

Take-away remarks

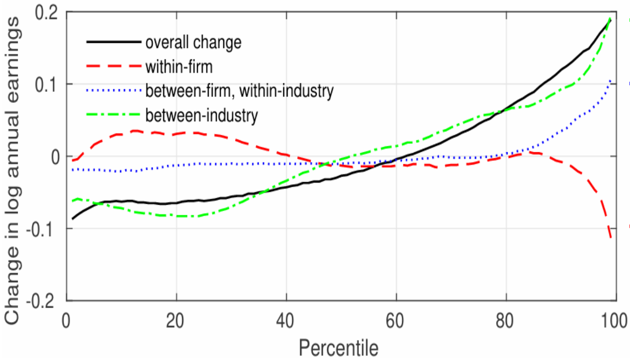
- Wage inequality in the U.S. economy since the 1990s has been substantial
  - dominant driver of increasing inequality is ***between-industry*** dispersion in wages

- Wage inequality is addressed by *technological heterogeneity* across industries
  - → primarily driven up by uneven, industry-level substitutability between workers;
  - → strengthen in sorting and segregation intensifies wage inequality;
  - → without its *structural effect*, SBTC does not address observed inequality.

- Moving beyond quantity-based narratives toward a focus on structural heterogeneity
  - what matters most is not the scale of factor input quantities accumulation, but the sector-specific evolution of technology and production structures

Appendix

It's the industry, not the firm



Contribution to wage inequality

- 61.9% between-industry;
- 23.1% between-firm, within-industry;
- 14.9% within-firm.

Source: Haltiwanger *et al.* (2024), period 1996-2018

Go back

This paper: empirics

Fact 1 (**Contribution**) *A small subset of industries drives the rise in wage inequality; these are in the tails of the industry-level wage growth distribution.*

Fact 2 (**Industry capital gaps**) *Dispersion in physical capital-labour ratio is decreasing across industries, while it is not that of ICT capital-labour ratio.*

Fact 3 (**Labour force composition**) *Increases in non-routine relative shares are determined by a substitution effect rather than by the employment size of industries.*

Fact 4 (**Dynamic transformations**) *Industries marked by highest changes in real wages experience a substantial rise in their non-routine workers relative share along an increasing ICT capital ratio dynamics.*

Go back

This paper: model

- ***Households***
  - task heterogeneity: *routine* and *non-routine* workers
  - endogenous sorting into firms and industries given heterogeneous productivities
  - it gives rise to imperfectly elastic labour supply $\leftrightarrow$ *sorting* and *segregation* effects
- ***Firms** and **Industries***
  - monopolistically competitive firms in each industry
  - two types of capital: *ICT* and *non-ICT* capital
  - non-routine labour is complementary to ICT capital; routine labour is substitutable

Go back

❇ Structural heterogeneity drives most of the between-industry real wage inequality

Findings:

① industry-specific shifts in elasticities among capital and workers are pivotal

- ➔ 94% if combined shifts in elasticities;
- ➔ 88% when considering also weights of factors of production.

② major labour market concentration intensifies wage inequality

- ➔ 98% if combined with joint shifts in elasticities;

③ these patterns hold with routine and non-routine workers separately

④ wage inequality is not altered by monopsony power in wage-setting

▸ Go back

Data from BEA

- Private non-residential capital types divided in gross categories[i]
  - each category contains different types of assets according to NIPA asset type code;
  - total of 96 different asset types.
- Classification:
  - (*i*) *digital equipment* is made of "Mainframes", "PCs", "DASDs", "Printers", "Terminals", "Tape Drives", "Storage Devices", and "System Integrators";
  - (*ii*) *intangible capital* coincides with "Total Intellectual Property Products" (IPP);
  - (*iii*) *ICT capital*, as the combination of *digital equipment* and *intangible capital*;
  - (*iv*) *physical* (or *non-ICT*) *capital* is the sum of all the remaining asset types.
- The value of each asset is expressed in millions of U.S. dollars

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[i] These are "Total Equipment", "Total Structures", and "Total Intellectual Property Products".

Data from BLS

- U.S. 2017 NAICS classification of industries only starting from 2003
- For each industry, occupations classified according to the U.S. SOC system
  - occupation based on the work performed and not on education or training
- I classify occupations by considering the "major" group:[ii]

  (*i*) *non-routine tasks* considers "Management", "Business and Financial Operations", "Computer and Mathematical", "Architecture and Engineering", "Life, Physical, and Social Science", "Community and Social Service", "Legal", "Educational Training and Library", and "Arts, Design, Entertainment, Sports, and Media" occupations;

  (*ii*) the rest of occupations is comprised in the set of *routine tasks*.

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[ii] These groups are a total of 22. I do not choose a more granular identification due to missing group definitions for some more detailed occupations.

Who drives increasing inequality?

- Industry $s \in \mathcal{S}$ contribution to wage inequality can be written as

$$\underbrace{\Delta\, var\left(w(s) - \overline{w}\right)}_{\substack{\text{between-industry}\\ \text{variance growth}}} = \sum_{s=1}^{S} \overbrace{\Delta \underbrace{\left(\frac{\ell(s)}{\ell}\right)}_{\substack{\text{employment}\\ \text{share}}} \underbrace{\left(w(s) - \overline{w}\right)^2}_{\substack{\text{relative}\\ \text{wage}}}}^{\substack{\text{industry-}s\text{ contribution to}\\ \text{between-industry variance}}} \tag{A.1}$$

- What portion of growth can be attributed to industry factors? *Shift-share analysis*

$$\underbrace{\Delta\left(\frac{\ell(s)}{\ell}\right)\left(w(s) - \overline{w}\right)^2}_{\substack{\text{industry-}s\text{ contribution to}\\ \text{between-industry variance}}} = \underbrace{\overline{\left(w(s) - \overline{w}\right)^2}\,\Delta\left(\frac{\ell(s)}{\ell}\right)}_{\text{shift share: employment}} + \underbrace{\overline{\left(\frac{\ell(s)}{\ell}\right)}\,\Delta\left(w(s) - \overline{w}\right)^2}_{\text{shift share: wage}} \tag{A.2}$$

→ relative importance of wage $(w(s))$ changes vs. employment share $(\ell(s))$ changes

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Estimates

Table A.1: Contribution to between-industry wage variance

				share	shift-share	
contribution	*industries*	*variance*		*employment*	*wage*	*employment*
$> 5\%$	3	.26	54%	8%	90%	10%
1% to 5%	7	.14	29%	18%	76%	24%
.05% to 1%	6	.05	10%	13%	124%	-24%
-.05% to .05%	46	.03	7%	61%	·	
wage growth						
bottom	15	.24	50%	31%	91%	9%
middle-bottom	16	.04	8%	28%	77%	23%
middle-top	16	.02	4%	15%	·	
top	15	.18	38%	26%	98%	2%

Estimates of eq. (A.1) for 3-digit U.S. 2017 NAICS industries. The last two columns quantify eq. (A.2) and, under "·", the shift-share for employment is highly less than zero. Operator Δ in the equations is $x_t - x_{t-1}$. Industries are grouped according to their contribution to between-industry wage inequality in the first part of the table while, under "wage growth", by the overall change in real log-wage. Source: BEA and own calculations.

▸ Go back

Decomposing total change in variance of real log-wages

Total rise in wage inequality can be written also as

$$
\begin{aligned}
\underbrace{\Delta\,\widetilde{var}\left(w_{s,t}-\overline{w}_t\right)}_{\text{total, wages}} &= \underbrace{\left(\frac{\ell_{g,0}}{\ell_0}\right)\left[\Delta\,\widetilde{var}\left(w_{s\in g,t}-\overline{w}_{g,t}\right)\right]}_{\text{within-group, wages}} + \underbrace{\sum_g\left[\Delta\,\widetilde{var}\left(\ell_{s,t}-\overline{\ell}_{g,t}\right)\right]\widetilde{var}\left(w_{s,0}-\overline{w}_{g,0}\right)}_{\text{between-groups, employment}} \\
&+ \underbrace{\sum_g\left[\Delta\,\widetilde{var}\left(w_{s,t}-\overline{w}_{g,t}\right)\right]\left[\Delta\,\widetilde{var}\left(\ell_{s,t}-\overline{\ell}_{g,t}\right)\right]}_{\text{between-groups, interaction}} \\
&+ \underbrace{\underbrace{\sum_{\mathcal{G}/g}\left(\frac{\ell_{\mathcal{G},0}}{\ell_0}\right)\left[\Delta\,\widetilde{var}\left(w_{s,t}-\overline{w}_{\mathcal{G},t}\right)\right]}_{\text{within-other groups, wages}} + \underbrace{\sum_g\left[\Delta\left(\frac{\ell_{g,t}}{\ell_t}\right)\widetilde{var}\left(\overline{w}_{g,t}-\overline{w}_t\right)\right]}_{\text{between groups, wages}}}_{\text{residual}}
\end{aligned}
\tag{A.3}
$$

where industries are partitioned in $g \in \mathcal{G}$ groups, and variances are *employment-weighted*. Estimates are presented in Table A.2

▸ Estimates

QUANTIFYING VARIANCE GROUP DECOMPOSITION

Table A.2: Decomposition of the rise in wage inequality

	industry group *g*				
share of the increase, wage variance	(1) *tails*	(2) *middle*	(3) *services*	(4) *manuf.*	(5) *other*
rising variance within the group	79%	32%	58%	11%	27%
employment reallocation across groups	34%	34%	17%	51%	10%
comovement (variance, employment)	7%	7%	3%	4%	5%
residual	-20%	27%	-22%	-34%	58%
total change across all industries	100%	100%	100%	100%	100%

Estimates of each component in eq. (A.3) for 3-digit U.S. 2017 NAICS industries between 2003 and 2022 related to $\log w(s)$*. Operator* Δ *in the equation is* $x_t - x_{t-1}$*, and not a percentage change. The first row shows the share of total increase in variance due to rising variance in the group of industries; the second row shows the share due to changes in employment between that group and the other industries in the sample (employment reallocation), holding constant the change in variance in each group; the third row shows the share that is due to the cross-product of rising variance and rising employment share; the fourth row is so that the sum for each column is 100%. "tails" and "middle" are referred to overall percentage changes distribution in industry real wage per capita; "manuf." stands for manufacturing industries, while "other" does not consider services and manufacturing industries. Source: BEA and own calculations.*

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Dispersion in wage and capital, cross-industries

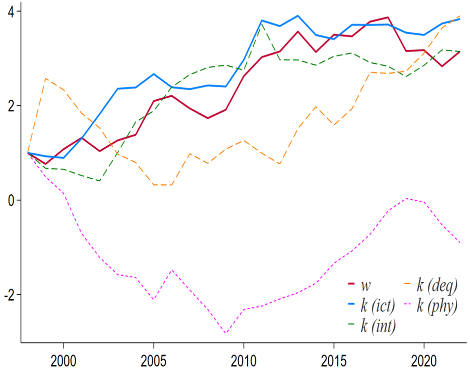

Figure A.1: Interquantile range, 90^{th}-10^{th}

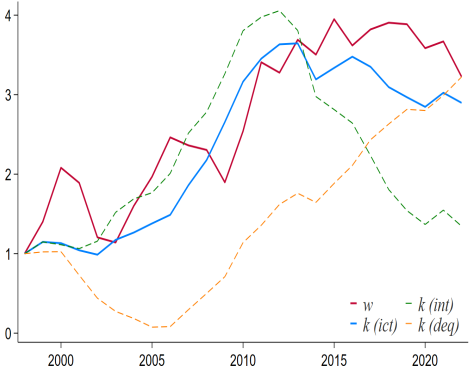

Figure A.2: Macro-complementarity at works

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Regressions (3-digit)

Table A.3: Regressions, capital stocks

	$log\, w\,(s)$			
	(1)	(2)	(3)	(4)
$\beta_{k^{phy}}$	.134*** (3.48)	.122** (2.94)	.164*** (9.05)	.114*** (3.52)
$\beta_{k^{ict}}$	.052* (2.00)			.049* (2.27)
$\beta_{k^{int}}$		.057* (2.03)		
$\beta_{k^{deq}}$		-.003 (-.26)		
$\beta_{k^{int}} \times \beta_{k^{deq}}$			.009** (2.99)	.009** (2.80)
R^2	.461	.491	.287	.497

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). t-statistics in parentheses. Analysis at 3/4-digit U.S. 2017 NAICS industries over 1998-2022 on* $N = 1650$ *observations. Fixed Effects (FE) regressions are of the form* $\log w_t\,(s) = \beta_C + \beta_i \mathcal{X}_{i,t} + u_t$*, with* $\mathcal{X}_i$ *representing the different capital-labour ratios considered. Results are robust when controlling for the log size of industries, or when taking capital series directly in levels. Variables are in log format. Constant not reported to save space. Source: BEA and own calculations.*

▸ Go back

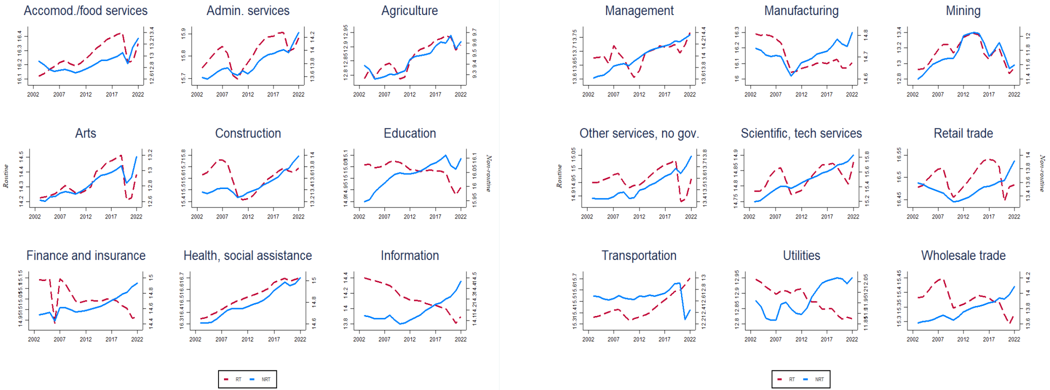
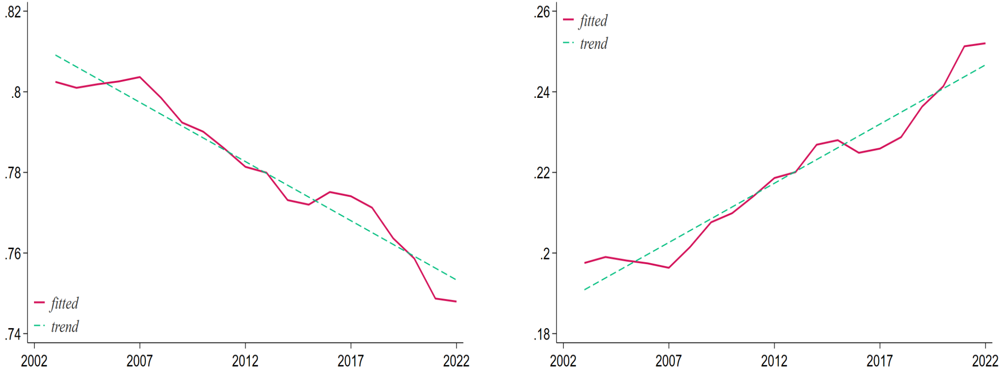


Figure A.3: Changes in routine and non-routine tasks

Go back



(a) Routine workers

(b) Non-routine workers

Figure A.4: Labour shares pattern

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Structural correlations

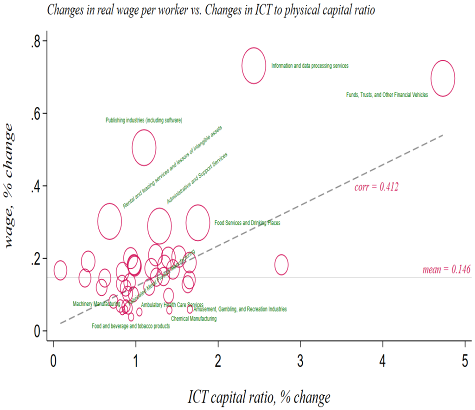

Figure A.5: Real wages vs. ICT capital ratio

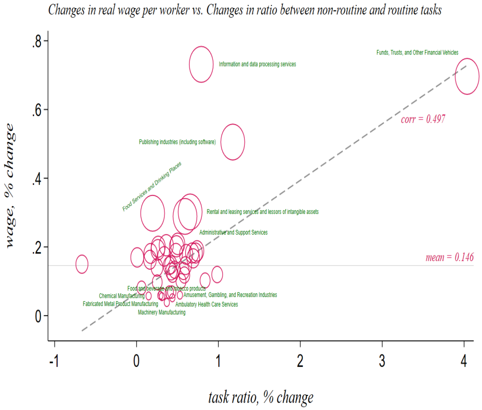

Figure A.6: Real wages vs. task ratio

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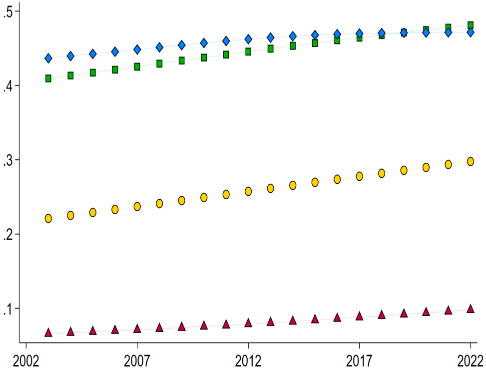
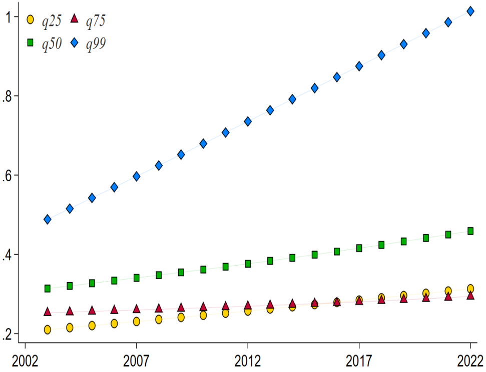
Trend by quantiles



(a) ICT capital ratio



(b) Task ratio

Figure A.7: Changes by percentiles

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Regressions, percentage changes

Table A.4: Combined regressions, percentage changes

	$\Delta\, log\, w\,(s)$		
	(1)	(2)	(3)
$\beta_{\Delta k}$	.054*** (.002)		.009 (.008)
$\beta_{\Delta \ell}$	.035* (.018)		.035*** (.007)
$\beta_{\Delta k \times \Delta \ell}$		.982 (.649)	.919*** (.152)
R^2	.030	.021	.037
N	1178	1178	1178

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). t-statistics in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries over 2003-2022. Fixed Effects (FE) regressions are of the form* $\Delta\, log\, w_t\,(s) = \beta_c + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$*, with* $\mathcal{X}_i$ *representing the percentage change in both ICT-to-physical capital and non-routine-over-routine workers ratios, and* $\mathcal{Z}_j$ *being a set of time-varying controls. Variables are in log format. Constant not reported to save space. Source: BEA, BLS and own calculations.*

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Frechéting-up the household problem

- Household-i of type-a in firm-h in industry-s owns an *indirect* utility

$$\mathcal{U}_h^i(a,s) = -\,log\left[\mathcal{I}^i\left(b^i, k^i(j), \mathcal{D}^i\right)\right] + log\left[w_h(a,s)\right] + \varsigma\, log\left[\frac{1}{g_h(a,s)}\right] + \wp_h^i(a,s) \quad \text{(B.1)}$$

- Idiosyncratic *productivity* in firm h, s is drawn once from a Frechét distribution

$$\mathcal{F}\left(\wp_{h,\ldots,H}^i(a,1), \ldots, \wp_{h,\ldots,H}^i(a,S)\right) = exp\left[-\sum_s \left(\int_h \wp_h^i(a,s)\, \mathrm{d}h\right)^{-\theta}\right]$$

- Upward-sloping labour supply curve of each a, h, s as in eq. (10)

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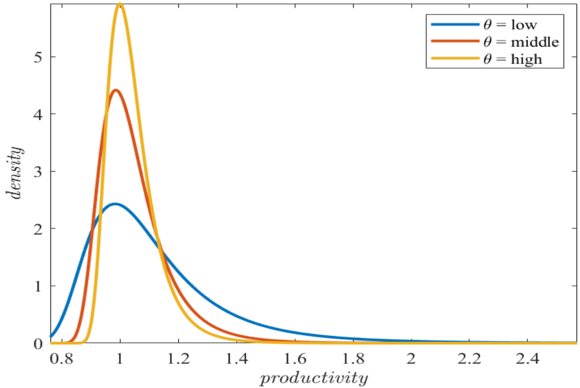

Figure B.1: Variability and the shape parameter

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Household inter-temporal problem

- The problem implied by eq. (B.1) can be rewritten inter-temporally as

$$\max_{\mathcal{C}_t^i,\, b_{t+1}^i,\, \{k_{t+1}^i(j)\}_{\forall j}} \mathcal{U}_{h,t}^i(a,s) = \sum_{t=0}^{\infty}\left[\beta^t \log \mathcal{C}_t^i\right] + \sum_{t=0}^{\infty}\left[\beta^t \log \mathcal{B}_{h,t}(a,s)\right] + \wp_h^i(a,s)$$

$$s.t. \quad \mathcal{C}_t^i + I_t^i(phy) + I_t^i(ict) + b_{t+1}^i - (1+r_t)\, b_t^i =$$

$$= w_{h,t}(a,s)\,\ell_h^i(a,s) + R_t\Big(k_t^i(phy) + k_t^i(ict)\Big) + \mathcal{D}_t^i$$

$$and \quad k_{t+1}^i(j) = \frac{I_t^i(j)}{\zeta_t^i} + (1-\delta_j)\, k_t^i(j) \quad , \quad \forall j = \{phy, ict\}$$

- Optimality conditions are in order

$$\frac{1}{\mathcal{C}_t^i} = \beta^t(1+r_{t+1})\frac{1}{\mathcal{C}_{t+1}^i}$$

$$R_t = (1+r_{t+1})\,\zeta_t^i - (1-\delta)\zeta_{t+1}^i \tag{B.2}$$

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Final output and industry demand

- A final producer competitively combines outputs from intermediate industries $s \in \mathcal{S}$

$$Y = \left(\int_0^1 y(s)^{\frac{\eta-1}{\eta}} \, \mathrm{d}s \right)^{\frac{\eta}{\eta-1}}$$

→ resulting demand of each industry s is $y(s) = \left(\frac{p(s)}{P}\right)^{-\eta} Y$, with the aggregate price index being $P = \left(\int_0^1 p(s)^{1-\eta} \, \mathrm{d}s \right)^{\frac{1}{1-\eta}} = 1$ since final good is competitively aggregated

- Industry-level output combines outputs from firms $h \in \mathcal{H}$ in that industry

$$y(s) = \left(\int_0^1 y_h(s)^{\frac{\epsilon-1}{\epsilon}} \, \mathrm{d}h \right)^{\frac{\epsilon}{\epsilon-1}}$$

→ resulting demand of each firm (h, s) is thus $y_h(s) = \left(\frac{p_h(s)}{p(s)}\right)^{-\epsilon} y(s)$, with $p(s) = \left(\int_0^1 p_h(s)^{1-\epsilon} \, \mathrm{d}h \right)^{\frac{1}{1-\epsilon}} = 1$ since industry-level good is competitively aggregated

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Production technology

- Firm-h in industry-s production function, $y = f\left(k^{phy}, k^{ict}, \ell^{rt}, \ell^{nrt}\right)$, is

$$y_h(s) = \left(k_h(phy,s)\right)^{\alpha}\Bigg\{\mu\left(\ell_h(rt,s)\right)^{\varsigma} + \\ + (1-\mu)\underbrace{\left[\lambda\left(k_h(ict,s)\right)^{\varrho} + (1-\lambda)\left(\ell_h(nrt,s)\right)^{\varrho}\right]^{\frac{\varsigma}{\varrho}}}_{q_h(s)}\Bigg\}^{\frac{1-\alpha}{\varsigma}} \tag{B.3}$$

- Elasticity indicators: $\varrho = \frac{\rho-1}{\rho}$ and $\varsigma = \frac{\sigma-1}{\sigma}$
  - capital-task complementarity requires that $\sigma > \rho$
  - parameters are all industry-specific

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Firm problem

- The problem of firm-h in industry-s is

$$\max_{p_h(s),\{k_h(j,s)\}_{\forall j},\{w_h(a,s)\}_{\forall a}} p_h(s)\, y_h(s) - \left(R \sum_j k_h(j,s) + \sum_a w_h(a,s)\ \ell_h(a,s) \right)$$

$$s.t. \quad y_h(s) = \left(\frac{p_h(s)}{p(s)} \right)^{\epsilon} y(s)$$

- Optimality conditions for capital types are in order

$$\begin{aligned} k_h(phy,s) &: \ p_h(s)\, F_{k_h(phy,s)} = \mathcal{M}\, R \\ k_h(ict,s) &: \ p_h(s)\, F_{k_h(ict,s)} = \mathcal{M}\, R \end{aligned} \tag{B.4}$$

$\longleftrightarrow$ pivotal to estimate the elasticities combination ρ, σ as given by eqs. (8.1)-(8.2)

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- Optimal wages for *routine* and *non-routine* workers in industry s are, respectively,

$$w(rt,s) = \left[\Lambda(s)\,\chi(rt,s)\,\left(k(phy,s)\right)^{\alpha}\,\mathcal{Y}^{\frac{1-\alpha-\varsigma}{\varsigma}}\,\mathcal{B}(rt,s)^{\theta(\varsigma-1)}\,\mathcal{WB}(rt)^{\theta(1-\varsigma)}\right]^{\frac{1}{1+\theta-\theta\varsigma}}$$

$$w(nrt,s) = \left[\Lambda(s)\,\chi(nrt,s)\,\left(k(phy,s)\right)^{\alpha}\,\mathcal{Y}^{\frac{1-\alpha-\varsigma}{\varsigma}}\,\mathcal{Q}^{\frac{\varsigma-\varrho}{\varrho}}\,\mathcal{B}(nrt,s)^{\theta(\varrho-1)}\,\mathcal{WB}(nrt)^{\theta(1-\varrho)}\right]^{\frac{1}{1+\theta-\theta\varrho}}$$

where $\chi(rt,s) = (1-\alpha)\mu$ and $\chi(nrt,s) = (1-\alpha)(1-\mu)(1-\lambda)$, $\Lambda(s) = p(s)\,\mathcal{M}^{-1}$, and $w(s) = (\mathcal{A})^{-1}\sum_a w(a,s)$

➔ Why do industry and firm levels coincide? See next

▸ Go back

▸ Validation: wage

Proposition 1 (Firm and industry layers)

In an economy characterized by sorting and segregation effects and a not perfectly elastic labour supply where the measure of workers in a given firm is determined by its wage relative to the others, as long as

(a) *firms within an industry have the same size; or*

(b) *workers are perfectly mobile across firms within an industry,*

$$\frac{\partial \ell_h(a,s)}{\partial w_h(a,s)} = -\frac{\partial \ell_{h'}(a,s)}{\partial w_h(a,s)} \quad \textit{and} \quad \frac{\partial w_h(a,s)}{\partial \ell_h(a,s)} = \frac{\partial w_{h'}(a,s)}{\partial \ell_{h'}(a,s)},$$

profit-maximizing wages set by firms in a specific industry are equal, and thus the unique optimal wage level can be directly written under industry notation. Moreover,

(c) *workers are immobile across firms between industries.*

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Why do industry and firm levels coincide?

Remark 1 (Max stability and workers' movement)
As from eq. (B.1), a worker has no incentive to choose a workplace where performing worse since

$$\mathcal{U}_h^i(a,s) \mid \wp_h^i(a,s)_{max} \gg \mathcal{U}_h^i(a,s) \mid \forall\, \wp_h^i(a,s) \in \left[\wp_h^i(a,s)_{min}\,,\, \wp_h^i(a,s)_{max}\right)$$

is ensured by each worker's selection of its maximal productivity for the (h,s) tuple. In addition, since firms are symmetric within an industry, the sorting choice is uniquely driven by the dispersion of households' productivities between industries so that workers are free to move across firms within an industry while getting the same utility.

▸ Go back

Aggregation

(a) *Labour market*

- aggregate labour supply is $L^S = \sum_a \sum_h \sum_s \ell_h(a,s)$
- aggregate labour demand is $L^D = \sum_a \sum_h \sum_s \ell_h(a,s)$ with $\ell_h(a,s) = \int_0^1 \ell_h^i(a,s)\,\mathrm{d}i$

(b) *Capital market* $\rightarrow K^D(phy) + K^D(ict) = K^S(phy) + K^S(ict)$

- capital types demands: $K^D(phy) = \sum_h \sum_s k_h(phy,s)$ and $K^D(ict) = \sum_h \sum_s k_h(ict,s)$
- capital types supplies: $K^S(phy) = \int_i k^i(phy)\,\mathrm{d}i$ and $K^S(ict) = \int_i k^i(ict)\,\mathrm{d}i$

(c) Aggregating the households' inter-temporal budget constraints and imposing the clearing conditions jointly with total quantities, the *aggregate resource constraint* at time t reads, given $\mathcal{B}_t = 1$, as

$$\mathcal{C}_t + I_t(phy) + I_t(ict) + b_{t+1} - (1+r_t)b_t = w_t\mathcal{B}_t L_t + R_t\Big(K_t(phy) + K_t(ict)\Big) + \mathcal{D}_t$$

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Equilibrium conditions

An equilibrium for this economy is defined as a households' choice of tasks, a combination of factors' prices, $\left(w\left(a,s\right),R\right)$, and a set of aggregate quantities, $\Omega=\left(Y,K\left(phy\right),K\left(ict\right),L\left(rt\right),L\left(nrt\right)\right)$ such that

- (a) each household picks the firm-industry tuple that maximizes eq. (B.1);
- (b) according to the occupational choice, each household maximizes its expected-utility version of the utility in eq. (B.1);
- (c) final and sectoral good producers maximize their revenues;
- (d) given the availability of workers in each job task as in eq. (10), optimal wages are determined by the equilibrium of labour demand and supply;
- (e) firms choose also capital bundles to maximize their profits;
- (f) all markets clear, shaping $\Omega\left(\cdot\right)$.

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Validating the wage equation

- Relevance of labour force composition on the industry-specific wage level

Table C.1: Industry wage and relative task size

	$\log w(s)$					
	(1)		(2)		(3)	
$g(rt/nrt, s)$	.016***	(.006)	.001	(.007)	-.020***	(.002)
$g(nrt/rt, s)$	.129***	(.036)	.250***	(.084)	.318***	(.043)
Industry FE	✓		✓		✗	
Time FE	✗		✓		✗	

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on N = 1240 observations. The Fixed Effects (FE) regressions are of the form* $\log w_t(s) = \beta_C + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $\mathcal{X}_i$ *being the regressors, and* $\mathcal{Z}_j$ *a set of controls. All series are in logs. Constant not reported to save space. Source: BEA, BLS and own calculations.*

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Validating the labour supply equation

- Relevance of relative wages on the industry-specific measures of worker types

Table C.2: Employment measures and tasks relative wages

	$\log \ell\,(rt,s)$			$\log \ell\,(nrt,s)$		
	(1)	(2)	(3)	(1)	(2)	(3)
$\ell(rt,s\,\vert\,w\mathcal{B})$	.765*	.657*	3.55***			
	(.32)	(.28)	(.17)			
$\ell(nrt,s\,\vert\,w\mathcal{B})$				5.76***	3.71***	29.4***
				(1.9)	(1.1)	(2.2)
Industry FE	✓	✓	✗	✓	✓	✗
Time FE	✗	✓	✗	✗	✓	✗

*Significance level at * (p<0.05), ** (p<0.01), *** (p<0.001). Standard error in parentheses. Analysis at 3-digit U.S. 2017 NAICS industries in 2003-2022 on* $N = 1240$ *observations. All the regressions are of the form* $y_t = \beta_C + \beta_i \mathcal{X}_{i,t} + \delta_j \mathcal{Z}_{j,t} + u_t$, *with* $\mathcal{X}_i$ *being the regressors, and* $\mathcal{Z}_j$ *a set of controls. All series are in logs. Constant not reported to save space. Source: BLS and own calculations.*

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Derivation of estimating equations

- Method of Karabarbounis and Neiman (2014). Steps:

1. define a CES production function, $y(\cdot)$ and compute the related F.O.C.s.;
2. define the following *income shares*. For a given labour force (ℓ), a given capital stock (k), and given profits $(\mathcal{D})$,

$$\phi_\ell = \left(\frac{1}{\mathcal{M}^\epsilon}\right)\left(\frac{w(\ell)\,\ell}{w(\ell)\,\ell + R\,k}\right), \ \phi_k = \left(\frac{1}{\mathcal{M}^\epsilon}\right)\left(\frac{R\,k}{w(\ell)\,\ell + R\,k}\right), \ \phi_{\mathcal{D}} = 1 - \frac{1}{\mathcal{M}^\epsilon};$$

3. by combining the F.O.C. for capital (either for labour) with all the above shares, one gets an equation whose left-hand side is $1 - \phi_\ell \mathcal{M}^\epsilon$. Then, this should be written in changes between two arbitrary periods, whose resulting elements should be "hatted", that is $\widehat{\cdot}$;
4. use eq. (B.2) to substitute $\widehat{R}$;
   - ⋆ from the Euler get $\widehat{(1+r)} = \frac{1}{\beta}$ so that, under constant β and δ, it holds that $\widehat{R} = \widehat{\zeta}$;
5. once substituting out $\widehat{R}$, take a linear approximation of the resulting equation around $\widehat{\zeta} = 0$, thereby obtaining the estimating equation.

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Table C.3: Baseline estimation

	β_ζ^ρ	*Std.Err.*	*95% CI*	β_ζ^σ	*Std.Err.*	*95% CI*	*Ind.*
bottom	.329	.04	[.250, .407]	.634	.08	[.482, .785]	16
middle	.420	.02	[.375, .466]	.400	.05	[.310, .491]	31
top	.249	.03	[.188, .310]	.766	.06	[.656, .877]	15

Estimation of the elasticities of substitution from eqs. (8.1)-(8.2), for 3-digit U.S. 2017 NAICS industries over the period 2003-2022. ρ refers to the degree of substitutability between non-routine workers and ICT capital; σ relates to the degree of substitutability between routine and non-routine workers.

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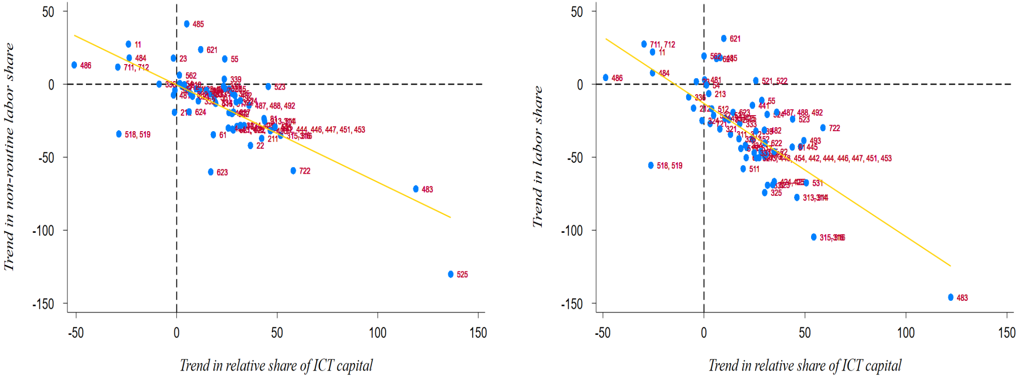
Negative relationship



(a) Estimates for ρ, $corr_\rho$ = -0.76

(b) Estimates for σ, $corr_\sigma$ = -0.75

Figure C.1: Correlation between elasticities and relative ICT capital

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Estimated elasticities, time-varying

Table C.4: Time-varying estimation

	2003-2012		2013-2022		
	β^{ρ}_{ζ}	β^{σ}_{ζ}	β^{ρ}_{ζ}	β^{σ}_{ζ}	*ind.*
bottom	.355 [.12]	.366 [.12]	.819 [.05]	.326 [.06]	16
middle	.431 [.04]	.429 [.08]	.345 [.04]	.438 [.09]	31
top	.408 [.04]	.367 [.12]	.508 [.06]	.358 [.05]	15

Estimation of the elasticities of substitution in eqs. (8.1)-(8.2) over different time spans (2003-2012 and 2013-2022), in absolute values, for 3-digit U.S. 2017 NAICS industries. Standard errors in parenthesis, and 95% confidence interval significant but not reported. ρ refers to the degree of substitutability between non-routine workers and ICT capital; σ relates to the degree of substitutability between routine and non-routine workers.

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Targeting moments for MSM

① Share parameters, λ and μ

- λ is matched with the industry-specific ICT capital in the aggregate stock;
- the weight of routine workers, μ, is used to bridge the share of routine workers in the data with that predicted by the model, *i.e.*, I implement the following identity

$$\ell_{model}(a,s) = \left(\frac{w(a,s)}{\mathcal{W}(a)}\,\frac{\mathcal{B}(a,s)}{\mathcal{B}(a)}\right)^{\theta} \approx \ell_{data}(a,s)$$

② Households' productivities dispersion parameter, θ

- it directly relates to the *between-industry* wage difference for worker-a: *wage premium* of type-rt working in *top* industry-s relative to its counterpart in the *bottom* industry-s':

$$\frac{w(rt,s)}{w(rt,s')} = \frac{\left[\Lambda(s)\,\chi(rt,s)\,\left(k(phy,s)\right)^{\alpha(s)}\,\mathcal{V}(s)^{\frac{1-\alpha(s)-\varsigma(s)}{\varsigma(s)}}\,\mathcal{B}(rt,s)^{\theta(\varsigma(s)-1)}\,\mathcal{WB}(rt)^{\theta(1-\varsigma(s))}\right]^{\frac{1}{1+\theta-\theta\varsigma(s)}}}{\left[\Lambda(s')\,\chi(rt,s')\,\left(k(phy,s')\right)^{\alpha(s')}\,\mathcal{V}(s')^{\frac{1-\alpha(s')-\varsigma(s')}{\varsigma(s')}}\,\mathcal{B}(rt,s')^{\theta(\varsigma(s')-1)}\,\mathcal{WB}(rt)^{\theta(1-\varsigma(s'))}\right]^{\frac{1}{1+\theta-\theta\varsigma(s')}}}$$

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MSM ESTIMATES

- Vector $\tilde{\Theta} = \{\lambda_s \,, \mu_s \,, \theta\}$ to match key moments of the production structure

$\longleftrightarrow$ minimize the *loss function* $\mathcal{L}\left(\tilde{\Theta}\right) = \left(\hat{m}\left(\tilde{\Theta}\right) - \tilde{m}\right)' \mathbf{W} \left(\hat{m}\left(\tilde{\Theta}\right) - \tilde{m}\right)$ to estimate $\tilde{\Theta}$

Table C.5: Method of Simulated Moments, results

	parameter	*value*	***moment to match***	*fit*	
				data	*model*
μ_{bot}	weight of routines in $y(bot)$	0.6763	routine share, bottom	.0858	.0434
μ_{mid}	weight of routines in $y(mid)$	0.4953	routine share, middle	.1357	.0458
μ_{top}	weight of routines in $y(top)$	0.3366	routine share, top	.0985	.0199
λ_{bot}	weight of ICT in $\mathcal{Q}(bot)$	0.4565	ICT share, bottom	.3968	.3968
λ_{mid}	weight of ICT in $\mathcal{Q}(mid)$	0.4645	ICT share, middle	.3042	.3042
λ_{top}	weight of ICT in $\mathcal{Q}(top)$	0.4514	ICT share, top	.2990	.2990
θ	productivity dispersion	11.302	wage premium, $w\left(a, [s, s']\right)$	.9945	.9945

Estimated values and related matched moment using the Methods of Simulated Moments.

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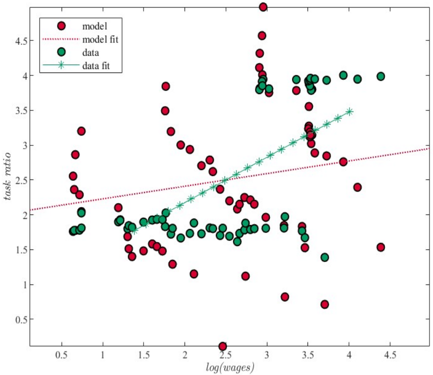
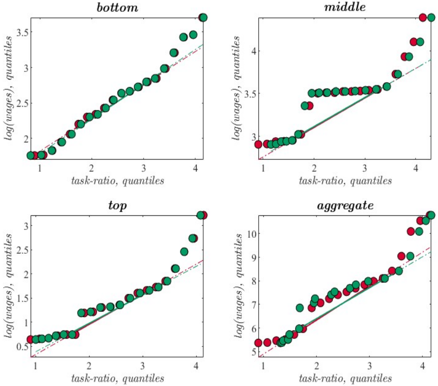


(a) Task Ratio vs. Wages



(b) Quantiles

Figure C.2: Model and data comparison (1/2)

Steady-state discussion: untargeted moments

Table C.6: Model fit, untargeted moments

	fit	
moment	*data*	*model*
aggregate task-premium	.001	.005
aggregate wage	-.008	-.073
routine wage, bottom	-.016	-.070
routine wage, middle	-.001	-.051
routine wage, top	-.007	-.079
non-routine wage, bottom	-.019	-.060
non-routine wage, middle	.003	-.074
non-routine wage, top	-.010	-.046

Untargeted moments to match to validate the calibration strategy. All moments, referred to real log-wages, are taken as percentage changes throughout the series.

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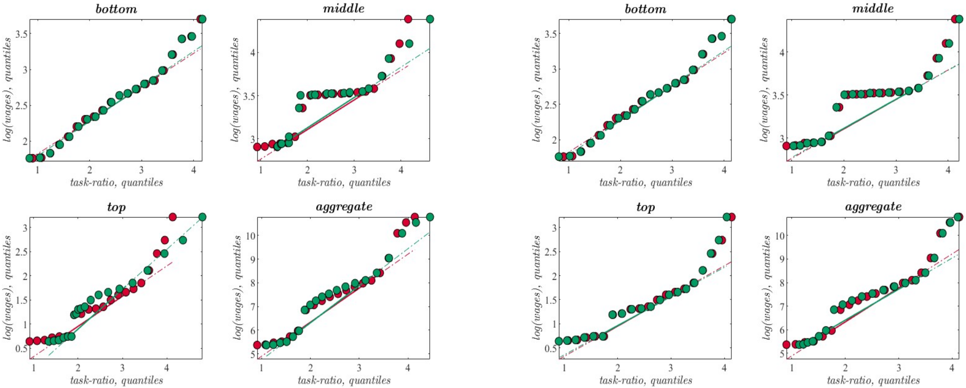
Steady-state quantiles



(a) Routines, quantiles

(b) Non-routines, quantiles

Figure C.3: Model and data comparison (1/2)

Herfindahl-Hirschman Index (HHI)

- The *Herfindahl-Hirschman Index* (HHI) is computed to account for market concentration of labour force at the industry level
- Market-level concentration for task-a is defined as

$$HHI_{\ell(a)} = \sum_{s_{|g}} \left(\frac{\ell(a, s_{|g})}{\ell(a)} \right)^2$$

where the sum is over individual industries (s), or over a group of industries $\left(s_{|g}\right)$

- Total employment concentration is defined by $HHI_{\ell} = \sum_a HHI_{\ell(a)}$
- The index is $HHI \in [0, 1]$:
  - a value of 1 identifies maximum market concentration, a single monopsonist;
  - conversely, a value of 0 results in a perfectly competitive environment.

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Herfindahl-Hirschman Index (HHI) by groups of industries

(a) Bottom　　**(b)** Middle　　**(c)** Top

Figure C.4: Market concentration by groups of industries

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Table C.7: Method of Simulated Moments, sub-periods

		2003-2012			2013-2022		
	moment to match	*value*	*data*	*model*	*value*	*data*	*model*
μ_{bot}	routine share, bottom	0.196	0.087	0.097	0.415	0.085	0.051
μ_{mid}	routine share, middle	0.486	0.132	0.057	0.506	0.140	0.030
μ_{top}	routine share, top	0.902	0.100	0.013	0.608	0.097	0.083
λ_{bot}	ICT share, bottom	0.650	0.399	0.399	0.786	0.395	0.395
λ_{mid}	ICT share, middle	0.530	0.314	0.314	0.490	0.295	0.295
λ_{top}	ICT share, top	0.185	0.287	0.287	0.327	0.311	0.311
θ	wage premium, $w\left(a,[s,s']\right)$	7.259	0.994	0.994	7.787	0.995	0.995

Estimated values and related matched moment using the Methods of Simulated Moments for the first and the second half of the sample.

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Models for counterfactual analysis

① Changing main parameters, fixing the others, and series over time

$$\Delta var\left(w\left(s\right)-\overline{w}\right)=f\left(\Phi_s\left(\tau_1\right),\Theta_s(\tau_1),\Phi_s\left(\tau_2\right)\mid\Delta\Theta_s\left(\Psi_{\tau_2},-\Psi_{\tau_1}\right)\right)\tag{Model A}$$

② Fixing main parameters, changing the others, and series over time

$$\Delta var\left(w\left(s\right)-\overline{w}\right)=f\left(\Phi_s\left(\tau_1\right),\Theta_s\left(\tau_1\right),\Phi_s\left(\tau_2\right)\mid\Delta\Theta_s\left(\Psi_{\tau_1},-\Psi_{\tau_2}\right)\right)\tag{Model B}$$

③ Impose same parameters, fixing them, and changing one or more series over time

$$var\left(w\left(s\right)-\overline{w}\right)=f\left(\Delta\Phi_s\left(x\right),\Phi_s\left(-x,\tau_0\right)\mid\Theta_{=\forall s}\right)\tag{Model C}$$

④ Fixing all parameters, and changing one or more series over time

$$\Delta var\left(w\left(s\right)-\overline{w}\right)=f\left(\Phi_s\left(x,\tau_1\right),\Delta\Phi_s\left(x_{\tau_2},-x_{\tau_1}\right)\mid\Theta_s\left(\tau_1\right)\right)\tag{Model D}$$

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Model re-calibration counterfactual, change

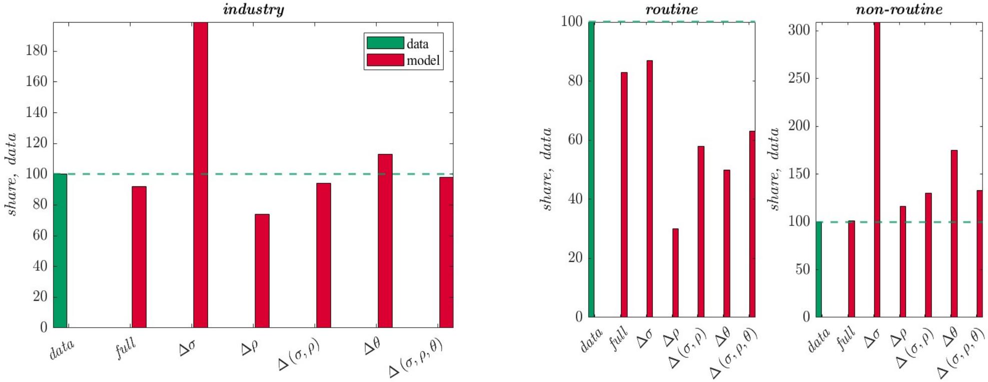

Figure C.5: Impact on between-industry wage inequality

▸ Go back

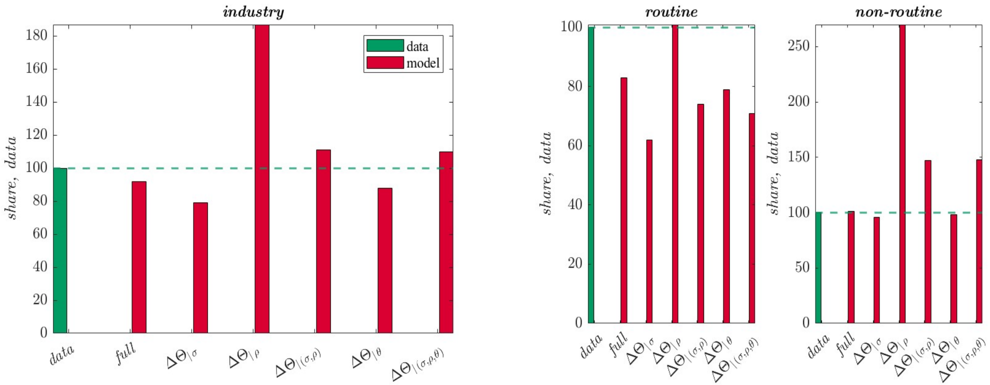

Figure C.6: Impact on between-industry wage inequality

▸ Go back

Model re-calibration counterfactual, change

- Between-industry real log-wage variance if only a specific parameter changes from period 1 to period 2, keeping the remaining parameters fixed at their initial level

Table C.8: Model counterfactual, change

		Δ ***model*** $\mid \Delta\, m\left(\Phi\left(\tau_2\right), \Theta = \left\{\Psi_{\tau_2}, -\Psi_{\tau_1}\right\}\right)$		
industry wages	*var* $(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.18			
Model	1.09			
$\Delta\sigma$		2.37	2.18	1.99
$\Delta\rho$		.88	.81	.74
$\Delta(\sigma, \rho)$		1.12	1.03	.94
$\Delta\theta$		1.34	1.23	1.13
$\Delta(\sigma, \rho, \theta)$		1.16	1.07	.98

Quantification of Model A. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to the aggregate economy considering bottom, middle, and top industries' groups.

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Model re-calibration counterfactual, fixing

- Between-industry real log-wage variance if only a specific parameter is fixed at its initial level, letting the remaining parameters to change from period 1 to period 2

Table C.9: Model counterfactual, fixing

industry wages	$var\,(w)_{\tau_2}$	Δ ***model*** $\vert\, \Delta\, m\left(\Phi\,(\tau_2)\,, \Theta = \{\Psi_{\tau_1}, -\Psi_{\tau_2}\}\right)$ *level*	*share, model*	*share, data*
Data	1.18			
Model	1.09			
$\Delta\Theta_{\vert\sigma}$		.94	.86	.79
$\Delta\Theta_{\vert\rho}$		2.21	2.04	1.87
$\Delta\Theta_{\vert(\sigma,\rho)}$		1.31	1.21	1.11
$\Delta\Theta_{\vert\theta}$		1.04	.96	.88
$\Delta\Theta_{\vert(\sigma,\rho,\theta)}$		1.30	1.20	1.10

Quantification of Model B. Model implied between-industry real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to the aggregate economy considering bottom, middle, and top industries' groups.

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Model re-calibration counterfactual, routine change

- Between-industry routine real log-wage variance if only a specific parameter changes from period 1 to period 2, keeping the remaining parameters fixed at their initial level

Table C.10: Model counterfactual, routine change

routine wages	*var* $(w)_{\tau_2}$	Δ ***model*** $\mid \Delta\, m\left(\Phi(\tau_2), \Theta = \{\Psi_{\tau_2}, -\Psi_{\tau_1}\}\right)$ *level*	*share, model*	*share, data*
Data	1.14			
Model	.95			
$\Delta\sigma$		.99	1.04	.87
$\Delta\rho$		.35	.37	.30
$\Delta(\sigma, \rho)$		.66	.70	.58
$\Delta\theta$		.57	.60	.50
$\Delta(\sigma, \rho, \theta)$		.71	.76	.63

Quantification of Model A. Model implied between-industry routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to the aggregate economy considering bottom, middle, and top industries' groups.

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Model re-calibration counterfactual, routine fixing

- Between-industry routine real log-wage variance if only a specific parameter is fixed at its initial level, letting the remaining parameters to change from period 1 to period 2

Table C.11: Model counterfactual, routine fixing

		Δ ***model*** $\mid \Delta\, m\left(\Phi\left(\tau_2\right), \Theta = \left\{\Psi_{\tau_1}, -\Psi_{\tau_2}\right\}\right)$		
routine wages	*var* $(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.14			
Model	.95			
$\Delta\Theta_{\mid\sigma}$		.71	.75	.62
$\Delta\Theta_{\mid\rho}$		1.15	1.22	1.01
$\Delta\Theta_{\mid(\sigma,\rho)}$		.84	.89	.74
$\Delta\Theta_{\mid\theta}$		.90	.95	.79
$\Delta\Theta_{\mid(\sigma,\rho,\theta)}$		.81	.86	.71

Quantification of Model B. Model implied between-industry routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to the aggregate economy considering bottom, middle, and top industries' groups.

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Model re-calibration counterfactual, non-routine change

- Between-industry non-routine real log-wage variance if only a specific parameter changes from period 1 to period 2, keeping the remaining parameters fixed at their initial level

Table C.12: Model counterfactual, non-routine change

		Δ ***model*** $\vert\, \Delta\, m\left(\Phi\left(\tau_2\right), \Theta = \left\{\Psi_{\tau_2}, -\Psi_{\tau_1}\right\}\right)$		
non-routine wages	*var* $(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.21			
Model	1.22			
$\Delta\sigma$		3.74	3.06	3.09
$\Delta\rho$		1.41	1.15	1.16
$\Delta(\sigma,\rho)$		1.57	1.28	1.30
$\Delta\theta$		2.11	1.73	1.75
$\Delta(\sigma,\rho,\theta)$		1.61	1.31	1.33

Quantification of Model A. Model implied between-industry non-routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to the aggregate economy considering bottom, middle, and top industries' groups.

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Model re-calibration counterfactual, non-routine fixing

- Between-industry non-routine real log-wage variance if only a specific parameter is fixed at its initial level, letting the remaining parameters to change from period 1 to period 2

Table C.13: Model counterfactual, non-routine fixing

		Δ ***model*** $\vert \Delta\, m\left(\Phi\left(\tau_2\right), \Theta = \left\{\Psi_{\tau_1}, -\Psi_{\tau_2}\right\}\right)$		
non-routine wages	*var* $(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.21			
Model	1.22			
$\Delta\Theta_{\vert\sigma}$		1.16	.95	.96
$\Delta\Theta_{\vert\rho}$		3.27	2.67	2.70
$\Delta\Theta_{\vert(\sigma,\rho)}$		1.78	1.45	1.47
$\Delta\Theta_{\vert\theta}$		1.18	.97	.98
$\Delta\Theta_{\vert(\sigma,\rho,\theta)}$		1.79	1.46	1.48

Quantification of Model B. Model implied between-industry routine workers real log-wage variance changes between two time spans differently calibrated, and changes also according to variations in some parameters; values are referred to the aggregate economy considering bottom, middle, and top industries' groups.

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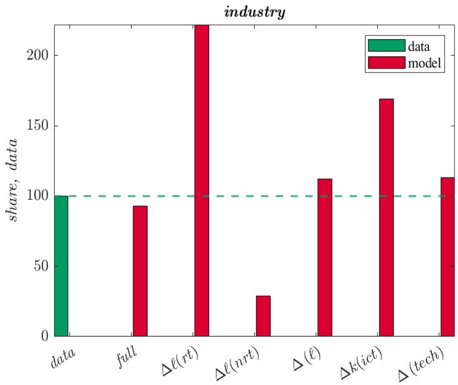
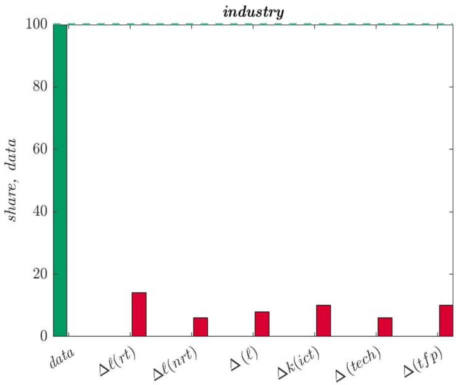
Model re-calibration counterfactual, SBTC and series



(a) Changing series, industry-specific calibration



(b) Changing series, unique calibration

Figure C.7: Impact on between-industry wage inequality

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Model re-calibration counterfactual, SBTC

- Between-industry real log-wage variance if only a specific series changes from period 1 to period 2, keeping the remaining parameters fixed at their initial level

Table C.14: Model counterfactual, SBTC

		Δ ***model*** $\mid \Delta\, m\left(\Phi = \{x_{\tau_2}, -x_{\tau_1}\} \middle| \Theta(\tau_1)\right)$		
industry wages	$var(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.18			
Model	1.09			
$\Delta\ell(rt)$		2.62	2.42	2.22
$\Delta\ell(nrt)$		.35	.32	.29
$\Delta\ell$		1.32	1.22	1.12
$\Delta k(ict)$		1.99	1.84	1.69
$\Delta(\ell, ict)$		1.34	1.23	1.13

Quantification of Model D. Model implied between-industry routine workers real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries.

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Model re-calibration counterfactual, routine SBTC

- Between-industry routine real log-wage variance if only a specific series changes from period 1 to period 2, keeping the remaining parameters fixed at their initial level

Table C.15: Model counterfactual, routine SBTC

		Δ ***model*** $\mid \Delta\ m\left(\Phi = \{x_{\tau_2}, -x_{\tau_1}\} \middle| \Theta(\tau_1)\right)$		
routine wages	$var(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.14			
Model	.95			
$\Delta\ell(rt)$		1.21	1.28	1.06
$\Delta\ell(nrt)$		.20	.21	.18
$\Delta\ell$		.56	.59	.49
$\Delta k(ict)$		.86	.91	.75
$\Delta(\ell, ict)$		.53	.56	.47

Quantification of Model D. Model implied between-industry routine workers real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries.

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Model re-calibration counterfactual, non-routine SBTC

- Between-industry non-routine real log-wage variance if only a specific series changes from period 1 to period 2, keeping the remaining parameters fixed at their initial level

Table C.16: Model counterfactual, non-routine SBTC

		Δ ***model*** $\mid \Delta\, m\left(\Phi = \{x_{\tau_2}, -x_{\tau_1}\} \,\middle|\, \Theta(\tau_1)\right)$		
non-routine wages	$var(w)_{\tau_2}$	*level*	*share, model*	*share, data*
Data	1.21			
Model	1.22			
$\Delta\ell(rt)$		4.04	3.30	3.34
$\Delta\ell(nrt)$		.50	.40	.41
$\Delta\ell$		2.09	1.70	1.73
$\Delta k(ict)$		3.13	2.56	2.59
$\Delta(\ell, ict)$		2.14	1.75	1.77

Quantification of Model D. Model implied between-industry non-routine workers real log-wage variance changes between two time spans uniformly calibrated, with changes according to variations in some series; values are referred to variance levels considering bottom, middle, and top industries.

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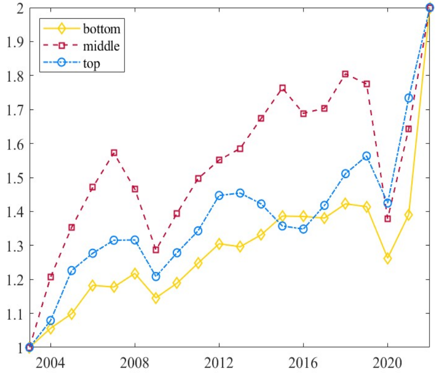
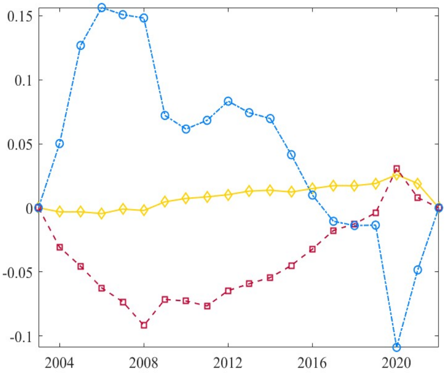


(a) Unique calibration, mean



(b) Difference with baseline

Figure C.8: Estimated productivities

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Model re-calibration counterfactual, series

- Between-industry real log-wage variance if only a specific series (or a combination) changes over time, keeping the remaining parameters fixed at their initial level, given the same (***mean***) calibration across industries

Table C.17: Model counterfactual, series

		model $\mid \Delta\,\Phi(x)$						
							$\Delta\,(tfp)$	
	data	$\Delta\ell(rt)$	$\Delta\ell(nrt)$	$\Delta\,(\ell)$	$\Delta k(ict)$	$\Delta\,(tech)$	*mean*	*baseline*
Wages, variance								
routine	2.285	.35	.16	.22	.24	.15	.25	.22
non-routine	2.314	.32	.12	.17	.23	.13	.21	.23
industry	2.299	.33	.14	.19	.23	.14	.23	.23

Quantification of Model C. Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the mean parameters to be homogeneous across industries.

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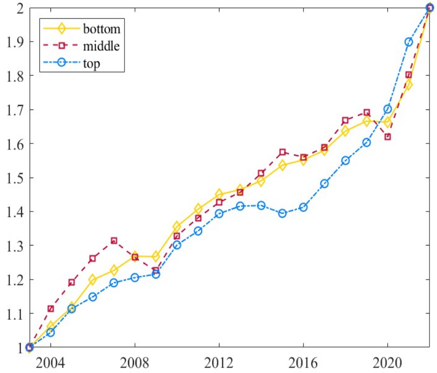
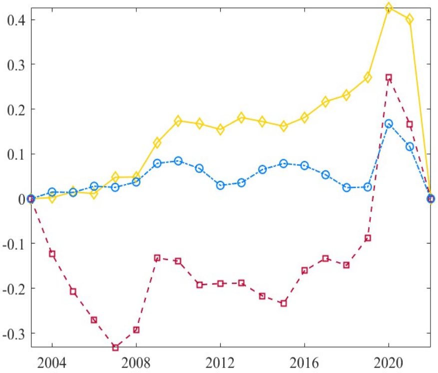


(a) Unique calibration, newly



(b) Difference with baseline

Figure C.9: Estimated productivities

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Model re-calibration counterfactual, series

- Between-industry real log-wage variance if only a specific series (or a combination) changes over time, keeping the remaining parameters fixed at their initial level, given the same (***new***) calibration across industries

Table C.18: Model counterfactual, series

		model $\mid \Delta\,\Phi(x)$						
							$\Delta\,(tfp)$	
	data	$\Delta\ell(rt)$	$\Delta\ell(nrt)$	$\Delta\ell$	$\Delta k(ict)$	$\Delta\,(\ell, ict)$	*newly*	*baseline*
Wages, variance								
routine	2.285	.62	.41	.46	.54	.40	.57	.34
non-routine	2.314	.07	.02	.02	.04	.03	.02	.09
industry	2.299	.35	.22	.24	.29	.21	.30	.22

Quantification of Model C. Changes in variances in real log-wages in the model induced by variations in one or more series, keeping fixed the others, and imposing the newly estimated parameters to be homogeneous across industries..

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